MORA

NOCA

Following the climate shift

← against the climate shift

with the climate shift →

Pinus albicaulis
Abies lasiocarpa
Tsuga mertensiana
Callitropsis nootkatensis
Picea engelmannii
Abies procera
Abies amabilis
Pinus monticola
Tsuga heterophylla
Pseudotsuga menziesii
Abies grandis
Pinus contorta
Thuja plicata
Pinus ponderosa

Closing the initial mismatch

← falls further behind

catches up with the lag →

Pinus albicaulis
Abies lasiocarpa
Tsuga mertensiana
Callitropsis nootkatensis
Picea engelmannii
Abies procera
Abies amabilis
Pinus monticola
Tsuga heterophylla
Pseudotsuga menziesii
Abies grandis
Pinus contorta
Thuja plicata
Pinus ponderosa

-0.10 -0.05 0.00 0.05 0.10 0.15

0.0 0.2 0.4 0.6

Species contribution

△ Holocene stability periods ○ Holocene climate-change periods ◇ Anthropocene